OBJECT ORIENTED SYSTEMS

Time: 1 hour | SET B | Full Marks: 30

1. a) Show with an example of a complete program how call by reference is implemented in Java.

b) State the difference between primitive and reference data type with example of each.

Discuss the functionality of *ensureCapacity()* method. State the difference between *length()* and *capacity()* methods with suitable examples. Explain the output of the following code snippet with proper logic.

```
StringBuffer s=new StringBuffer();
s.append("Information");
System.out.println(s.capacity());
s.ensureCapacity(40);
System.out.println(s.capacity());
```

c) Discuss the functionalities (with full function signatures and appropriate examples) of the following methods: *compareToIgnoreCase(), substring(int i, int j), indexOf(), replace()* and *insert()*.

[3+(2+2+3)+5=15]

2. a) Show with a complete program how a function can be overloaded in case of inheritance. What is the basic difference between function overloading and overriding?

b) Consider a case of multi-level inheritance where a class *Base* is inherited by the class *Child* which is again inherited by the class *GrandChild*. Suppose the class *Base* has a parameterized constructor and a method *display()*. How you can call them through an object of the class *GrandChild*. Show with complete program.

c) Assume a class Sample. Consider the following two scenarios of creating objects. Show diagrammatically how in memory the references point to the objects that are created for the two cases below.

```
Sample s1=new Sample();
Sample s2;
s2=s1;
Sample s3=new Sample();
s1=s3;
```

```
Sample s1=new Sample(), s2;
Sample s3=new Sample();
s2=s3;
s3=new Sample();
s2=s3;
```

[(3+1)+4+(1+3+3)=15]